EXPERIMENT 7

BANKER'S DEADLOCK AVOIDANCE ALGORITHM

AIM:

To implement the Banker's Algorithm for deadlock avoidance and determine whether the system is in a safe state.

PROCEDURE:

1. Initialize work=available and finish[i]=false for all values of i

2.Find an i such that both:

finish[i]=false and Needi<= work

3. If no such i exists go to step 6

4. Compute work=work+allocationi

5. Assign finish[i] to true and go to step 2

6. If finish[i]==true for all i, then print safe sequence

7. Else print there is no safe sequence

PROGRAM:

```
#include <stdio.h>
int main()
{
    // P0, P1, P2, P3, P4 are process names
    int n, m, i, j, k;
    n = 5; // Number of processes
    m = 3; // Number of resources
    // Allocation Matrix
    int alloc[5][3] = {
        {0, 1, 0},
        {2, 0, 0},
        {3, 0, 2},
        {2, 1, 1},
        {0, 0, 2}
    };

    // Maximum Matrix
    int max[5][3] = {
        {7, 5, 3},
        {3, 2, 2},
```

```
    {9, 0, 2},
    {2, 2, 2},
    {4, 3, 3}
};

// Available Resources
int avail[3] = {3, 3, 2};

int f[n], ans[n], ind = 0;

for(k = 0; k < n; k++)
{
    f[k] = 0;
}

int need[n][m];

for(i = 0; i < n; i++)
{
    for(j = 0; j < m; j++)
        need[i][j] = max[i][j] - alloc[i][j];
}

int y = 0;

for(k = 0; k < 5; k++)
{
    for(i = 0; i < n; i++)
    {
        if(f[i] == 0)
        {
            int flag = 0;

            for(j = 0; j < m; j++)
            {
                if(need[i][j] > avail[j])
                {
                    flag = 1;
                    break;
                }
            }

            if(flag == 0)
            {
                ans[ind++] = i;

                for(y = 0; y < m; y++)
                    avail[y] += alloc[i][y];

                f[i] = 1;
            }
```

```
            }
        }
    }

    printf("The SAFE Sequence is \n");

    for(i = 0; i < n - 1; i++)
        printf(" P%d ->", ans[i]);

    printf(" P%d", ans[n - 1]);

    return (0);
}
```

OUTPUT:



RESULT:

Thus, the Banker's Algorithm for deadlock avoidance was implemented successfully, and the system was found to be in a safe state.